

Let $X_1, \dots, X_N$ be independent sub-gaussian random variables with zero means and unit variances, and let $a = (a_1, \dots, a_N) \in \mathbb{R}^N$. Prove that for every $p \in [2, \infty)$ we have

$$\left(\sum_{i=1}^N a_i^2 \right)^{1/2} \leq \left\| \sum_{i=1}^N a_i X_i \right\|_{L^p} \leq CK \sqrt{p} \left(\sum_{i=1}^N a_i^2 \right)^{1/2}$$

where $K = \max_i \|X_i\|_{\psi_2}$ and C is an absolute constant.

$$\mathbb{E} \left[\left(\sum_{i=1}^N a_i^2 \right)^p \right]^{1/p}$$

L_p norm:

$$\|X\|_{L_p} = \mathbb{E} [|X|^p]^{1/p} \quad p \in [1, \infty)$$

$$L_p = \left\{ X \mid \|X\|_{L_p} < \infty \right\} \quad \text{normed space.}$$

$p=2 \Rightarrow$ inner product space.

$$\langle X, Y \rangle_{L_2} = \mathbb{E} X Y$$

$$\langle X, X \rangle = \mathbb{E}[X^2]$$

X is subgaussian:

$$1. \exists K_1 > 0: \text{ (Tail)} \quad \mathbb{P}(|X| \geq t) \leq 2 \exp\left(-\frac{t^2}{K_1^2}\right) \quad \text{for } t \geq 0$$

$$2. \exists K_2 > 0: \text{ (Moment)} \quad \|X\|_{L_p} \leq K_2 \sqrt{p} \quad \text{for } p \geq 1$$

$$3. \exists K_3 > 0: \text{ (MGF } X^2) \quad \mathbb{E}\left[e^{\frac{X^2}{K_3^2}}\right] \leq 2$$

(If X is zero-mean)

$$4. \exists K_4 > 0: \quad \mathbb{E}[e^{\lambda X}] \leq \exp(K_4^2 \lambda^2) \quad \lambda \in \mathbb{R}$$

$$\textcircled{1} \rightarrow \textcircled{2} \quad \times \rightarrow \left(\frac{x}{k} \right)$$

$$\|X\|_p^p = \int_0^\infty P(|X|^p \geq u) du$$

$$u = t^p$$

$$= \int_0^\infty \underbrace{P(|X| \geq t)}_{\textcircled{1}} p t^{p-1} dt$$

$$\leq \int_0^\infty \exp(-t^2/k^2) p t^{p-1} dt$$

$$\frac{t^2}{k^2} = s$$

$$= C \int_0^\infty \exp(-s) \underbrace{p k^{p-1}}_{\rightarrow} s^{\frac{p-1}{2}} ds \quad t = k\sqrt{s}$$

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \quad = \quad k \Gamma(ap + b)$$

$$\Gamma(z) \leq 3z^2$$

$$\textcircled{2} \rightarrow \textcircled{3}$$

$$K_2 \rightarrow \textcircled{1}$$

$$X \rightarrow \frac{X}{K_2}$$

$$\mathbb{E}[e^{X^2/K_2^2}] \leq 2$$

$$\mathbb{E}[e^{\lambda^2 X^2}] = 1 + \mathbb{E}\left[\sum_{i=1}^{\infty} \frac{(\lambda X)^{2i}}{i!}\right]$$

$$= 1 + \sum_{i=1}^{\infty} \frac{\mathbb{E}[(\lambda X)^{2i}]}{i!} \leq 1 + \sum_{i=1}^{\infty} \frac{e^i}{i!} \mathbb{E}[X^{2i}]$$

$$\left(\frac{n}{e}\right)^n \leq n! \leq en \left(\frac{n}{e}\right)^n \quad \left\{ \begin{array}{l} \mathbb{E}[|X|^{2p}] \leq (2p)^p \\ \leq 1 + \sum_{i=1}^{\infty} \frac{e^i}{i!} (2i)^i \\ = 1 + \sum (2e)^i \end{array} \right.$$

$$\lambda = \frac{1}{2\sqrt{e}}$$

$$= \frac{1}{1 - 2e\lambda^2} \leq 2$$

Orlicz Norm

$$\psi: \mathbb{R}_{\geq 0}^+ \rightarrow \mathbb{R}_{\geq 0}^+$$

ψ -Orlicz of X :

$$\|X\|_{\psi} = \inf \left\{ t > 0 \mid \mathbb{E} \left[\psi \left(\frac{|X|}{t} \right) \right] \leq 1 \right\}$$

e.g. $\psi(u) = u^p \Rightarrow L_p$ norm

strictly
convex increasing

$$\boxed{\psi(\bullet) \geq 0}$$



$$\psi(u) = e^{u^2} - 1 \quad \text{=: } \psi_q(u)$$

$$\underbrace{q=2}_{\psi(u) = e^{u^2} - 1} \quad \|X\|_{\psi_2} = \inf \left\{ t > 0 \mid \mathbb{E} \left[e^{\frac{X^2}{t^2}} - 1 \right] \leq 1 \right\}$$

$$= \inf \left\{ t > 0 \mid \mathbb{E} \left[e^{X^2/t^2} \right] \leq 2 \right\}$$

$$\|X\|_{\psi_2} \text{ finite} \iff X \text{ is subgaussian}$$

$$q=1 \quad \|X\|_{\psi_1} = \underline{\hspace{2cm}}$$

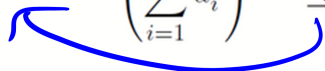
$$\|X\|_{\psi_1} \text{ finite} \iff X \text{ is subexponential}$$

$$1. \quad \mathbb{P}(|X| \geq t) \leq 2 \exp\left(-\frac{ct^2}{\|X\|_{\psi_2}^2}\right)$$

$$2. \quad \|X\|_{L_p} \leq C \|X\|_{\psi_2} \sqrt{p}$$

$$3. \quad \mathbb{E} \left[\frac{X^2}{\|X\|_{\psi_2}^2} \right] \leq 2$$

Let $X_1, \dots, X_N$ be independent sub-gaussian random variables with zero means and unit variances, and let $a = (a_1, \dots, a_N) \in \mathbb{R}^N$. Prove that for every $p \in [2, \infty)$ we have

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where $K = \max_i \|X_i\|_{\psi_2}$ and C is an absolute constant.

$$\| \sum X_i \|_{L_2} = \sum \|X_i\|_{L_2}$$

$$\| \sum X_i \|_{L_p} \leq \sum \|X_i\|_{L_p}$$

Let $X \in \mathbb{R}^d$ be a sub-Gaussian random vector with variance proxy σ^2 . Then

$$\mathbb{E} \left[\max_{\theta \in \mathcal{B}_2} \theta^\top X \right] = \mathbb{E} \left[\max_{\theta \in \mathcal{B}_2} |\theta^\top X| \right] \leq 4\sigma\sqrt{d}.$$

Moreover, for any $\delta > 0$, with probability $1 - \delta$, it holds

$$\max_{\theta \in \mathcal{B}_2} \theta^\top X = \max_{\theta \in \mathcal{B}_2} |\theta^\top X| \leq 4\sigma\sqrt{d} + 2\sigma\sqrt{2\log(1/\delta)}.$$

subgaussian R.V.

$$\lambda \in \mathbb{R}^d \Rightarrow \mathbb{E} \left[e^{\lambda^\top X} \right] \leq e^{\frac{\lambda^2 \sigma^2}{2}}$$
$$\lambda \in \mathbb{R}^d \Rightarrow \mathbb{E} \left[e^{\langle \lambda, X \rangle} \right] \leq e^{\frac{\|\lambda\|_2^2 \sigma^2}{2}}$$

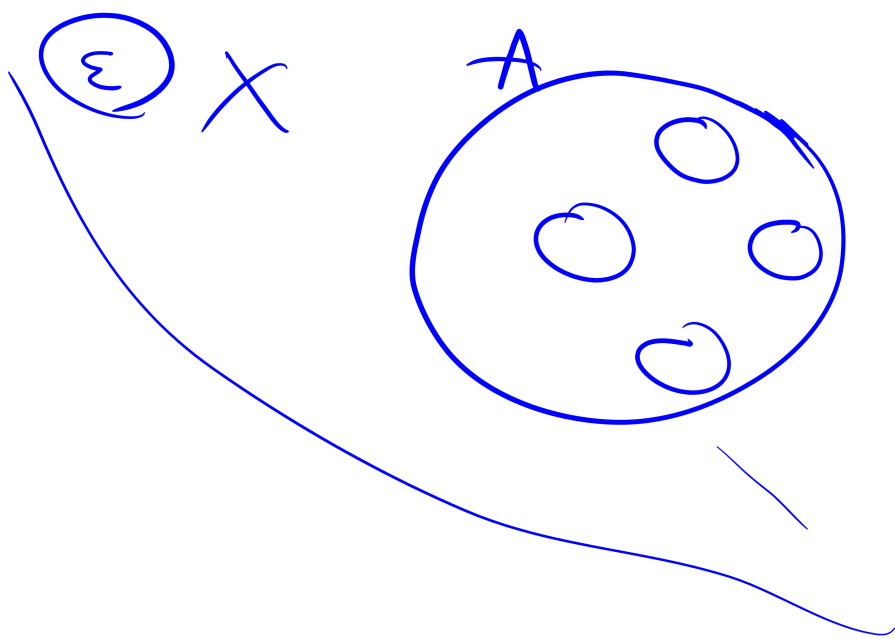
(X, d)

$$N \subseteq A \subseteq X$$

ε -net :

$$\forall x \in A \quad \exists n \in N : d(x, n) \leq \varepsilon$$

$N(\varepsilon, A, d)$ covering number.



N is ε -separated

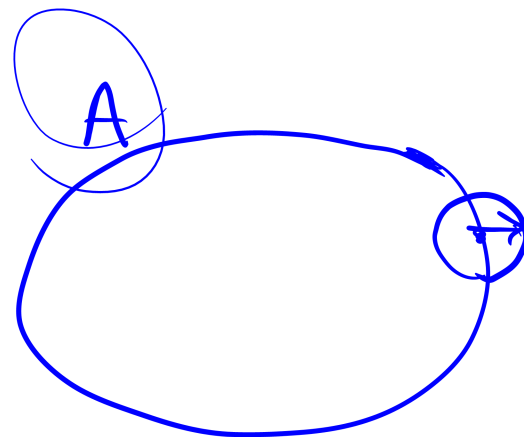
maximal ε -separated set. $\Rightarrow$ is a ε -net

$B(0,1) \subset \mathbb{R}^d$ $\text{Area}(r) = c_d r^d$

K is a ε -separated set.
maximal

$$N(\varepsilon) \cdot c_d (\varepsilon/2)^d \leq c_d (1 + \varepsilon/2)^d$$

$$|K| = N(\varepsilon) \leq (1 + 2/\varepsilon)^d \leftarrow$$



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Moreover, for any $\delta > 0$, with probability $1 - \delta$, it holds

$$\max_{\theta \in \mathcal{B}_2} \theta^\top X = \max_{\theta \in \mathcal{B}_2} |\theta^\top X| \leq \underbrace{4\sigma\sqrt{d} + 2\sigma\sqrt{2\log(1/\delta)}}_{\varepsilon}.$$

$$\varepsilon = \frac{1}{2}$$

$$|N| \leq 5^d$$

$$\|X\|_2 \leq 2 \max_{v \in N} \langle v, X \rangle$$

$$\mathbb{P}(\|X\| \geq t) \leq \underbrace{\hspace{10em}}$$

Let $X_1, \dots, X_n$ be an independent and identically distributed sequence taking values in $[m]$. For $i \in [m]$, let

$$p(i) = \mathbb{P}(X_1 = i) \quad \text{and} \quad \hat{p}(i) = \frac{1}{n} \sum_{t=1}^n \mathbb{I}\{X_t = i\}.$$

Show that for any $\delta \in (0, 1)$,

$$\mathbb{P}\left(\|p - \hat{p}\|_1 \geq \sqrt{\frac{2 \left[\log\left(\frac{1}{\delta}\right) + m \log(2)\right]}{n}}\right) \leq \delta.$$

$$(s_1, \dots, s_m) = s \quad s_i \in \{-1, +1\}$$

$$\|p - \hat{p}\|_1 = \max_s \langle s, p - \hat{p} \rangle$$

Union Bound over all possible vectors m .